Practical no. 18

Aim - Write a program to read the CPU%, Memory% and CPU temperature of Raspberry Pi.

Hardware Requirements

1. Raspberry Pi Model A/B/B+

- **Software Requirements**

  1. Raspbian Stretch OS

Code:-

```
#!/usr/bin/env python3

import psutil
import time

def get_cpu_temperature():
    """Read Raspberry Pi CPU temperature."""
    try:
        with open("/sys/class/thermal/thermal_zone0/temp", "r") as file:
            temp = int(file.read()) / 1000.0
        return temp
    except FileNotFoundError:
        return None

while True:
    # CPU usage percentage
    cpu_usage = psutil.cpu_percent(interval=1)

    # Memory usage percentage
    memory_usage = psutil.virtual_memory().percent

    # CPU temperature
    cpu_temp = get_cpu_temperature()

    print("---------------------------------")
    print(f"CPU Usage        : {cpu_usage:.1f}%")
    print(f"Memory Usage     : {memory_usage:.1f}%")

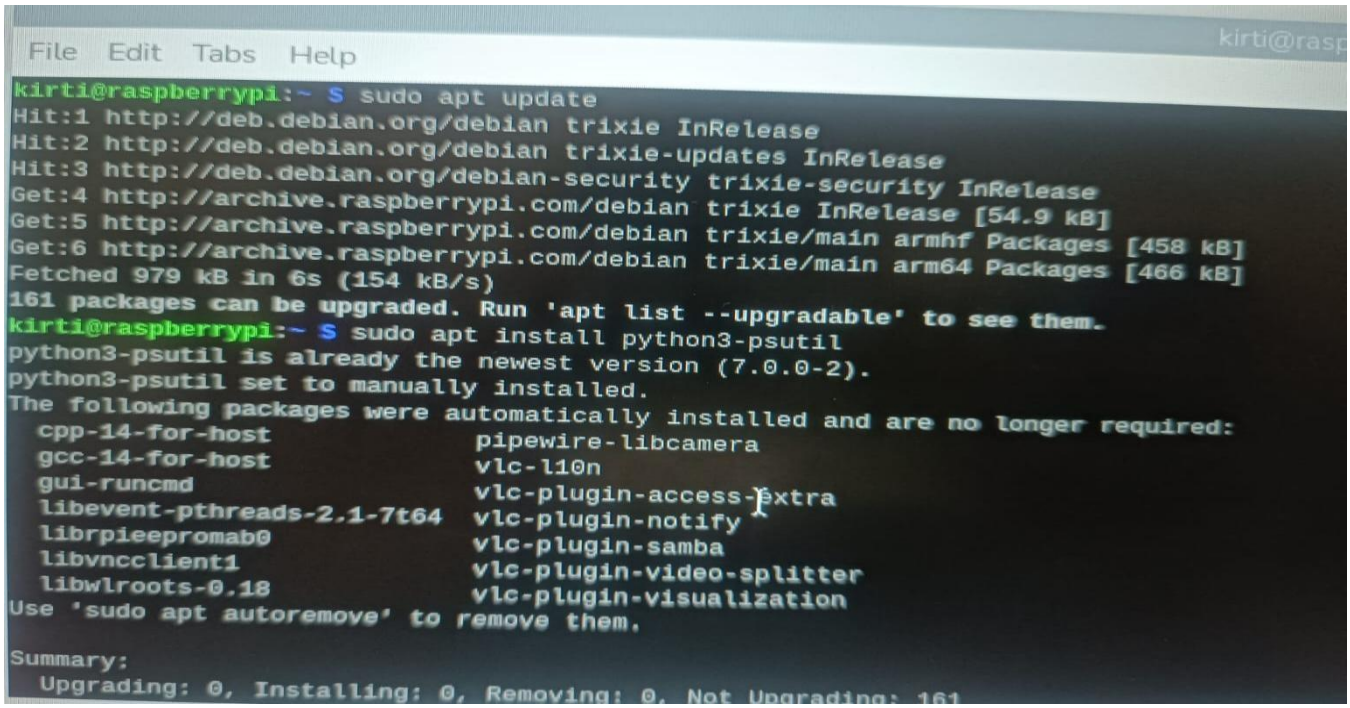
    if cpu_temp is not None:
        print(f"CPU Temperature : {cpu_temp:.1f} °C")
    else:
        print("CPU Temperature : Not available")

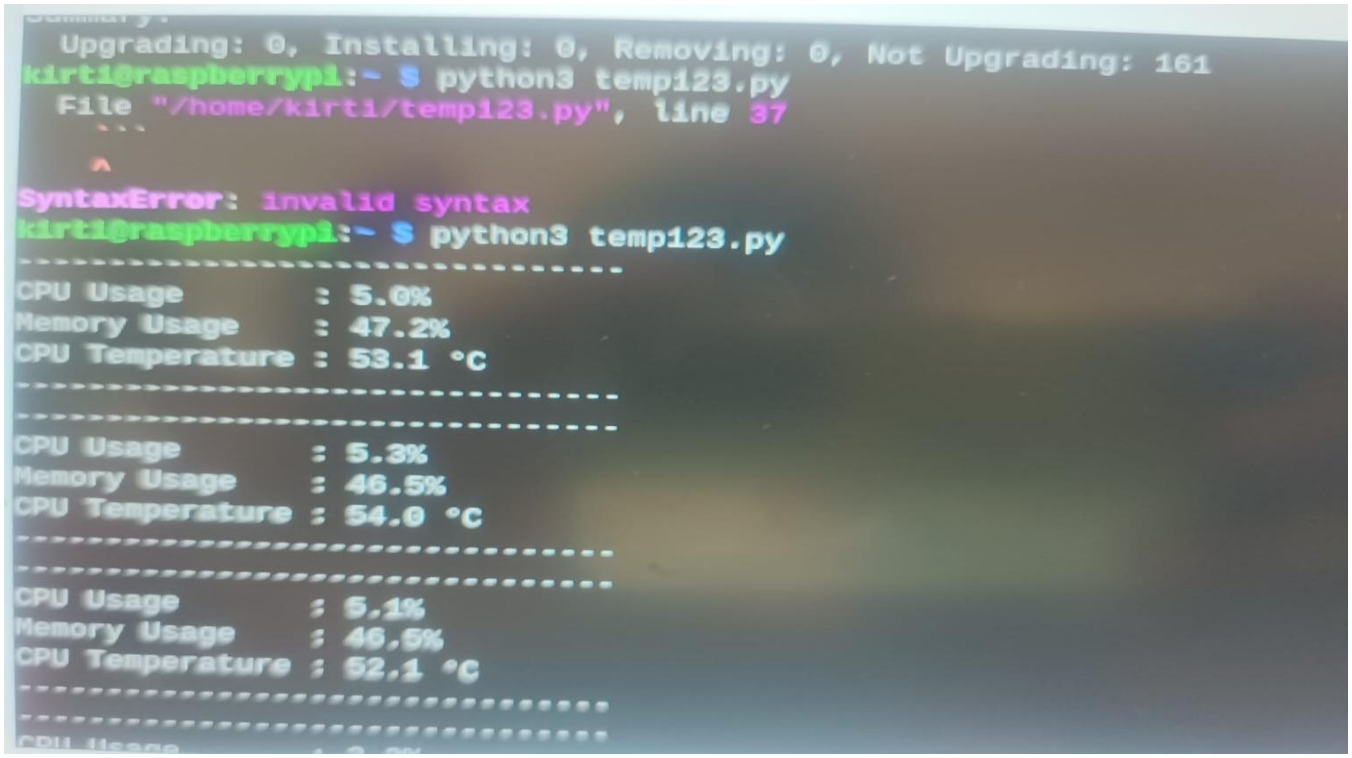
    print("---------------------------------")

    time.sleep(1)
```

To Run the Programme:-

1. **sudo apt update**
2. **sudo apt install python3-psutil**
3. **python3 test.py**


File Edit Tabs Help
kirti@rasp
kirti@raspberrypi:~ $ sudo apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Get:4 http://archive.raspberrypi.com/debian trixie InRelease [54.9 kB]
Get:5 http://archive.raspberrypi.com/debian trixie/main armhf Packages [458 kB]
Get:6 http://archive.raspberrypi.com/debian trixie/main arm64 Packages [466 kB]
Fetched 979 kB in 6s (154 kB/s)
161 packages can be upgraded. Run 'apt list --upgradable' to see them.
kirti@raspberrypi:~ $ sudo apt install python3-psutil
python3-psutil is already the newest version (7.0.0-2).
python3-psutil set to manually installed.
The following packages were automatically installed and are no longer required:
cpp-14-for-host
gcc-14-for-host
gui-runcmd
libevent-pthreads-2.1-7t64
librpieepromab0
libvncclient1
libwlroots-0.18
pipewire-libcamera
vlc-l10n
vlc-plugin-access-extra
vlc-plugin-notify
vlc-plugin-samba
vlc-plugin-video-splitter
vlc-plugin-visualization
Use 'sudo apt autoremove' to remove them.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 161


Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 161
kirti@raspberrypi:~ $ python3 temp123.py
File "/home/kirti/temp123.py", line 37
^
SyntaxError: invalid syntax
kirti@raspberrypi:~ $ python3 temp123.py
CPU Usage : 5.0%
Memory Usage : 47.2%
CPU Temperature : 53.1 °C
CPU Usage : 5.3%
Memory Usage : 46.5%
CPU Temperature : 54.0 °C
CPU Usage : 5.1%
Memory Usage : 46.5%
CPU Temperature : 52.1 °C